Schulich School of Engineering

ENEL 678 - Graduate Project in Electrical Engineering

UNIVERSITY OF CALGARY

Final Project Presentation

Topic:
Design and Optimization of a Smart EV Charging Station with Integrated Solar PV and Electrical Grid

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Presented by: TEAM CHARGED UP

Product Scope (Proposed)

- This project aims to build a smart EV charging station close to Jasper by combining solar energy, a battery energy storage system (BESS), and a power grid connection. An important step toward enhancing the sustainability and dependability of EV charging infrastructure, especially in rural areas, is this smart EV charging station project.
- The adoption of electric vehicles and the decrease in carbon emissions will both be greatly aided by this project. This project is a crucial component of the ongoing shift to clean and sustainable transportation because it integrates solar energy, battery storage, microgrid technologies, and DC fast charging capabilities.

Key Objective:

- Promote sustainable EV charging through solar energy integration.
- Enhance system reliability using battery storage and grid backup.
- Enable efficient, high-speed EV charging in remote, high-traffic locations.

Core:

- **PV System with MPPT Optimization**
- **Energy Source Synchronization**
- **Microgrid Compatibility**
- **Charging Circuit Simulation**
- **Advanced Software Control**
- **DC Fast Charging Capability**
- **Battery Energy Storage System (BESS)**

Product Deliverables (Proposed)

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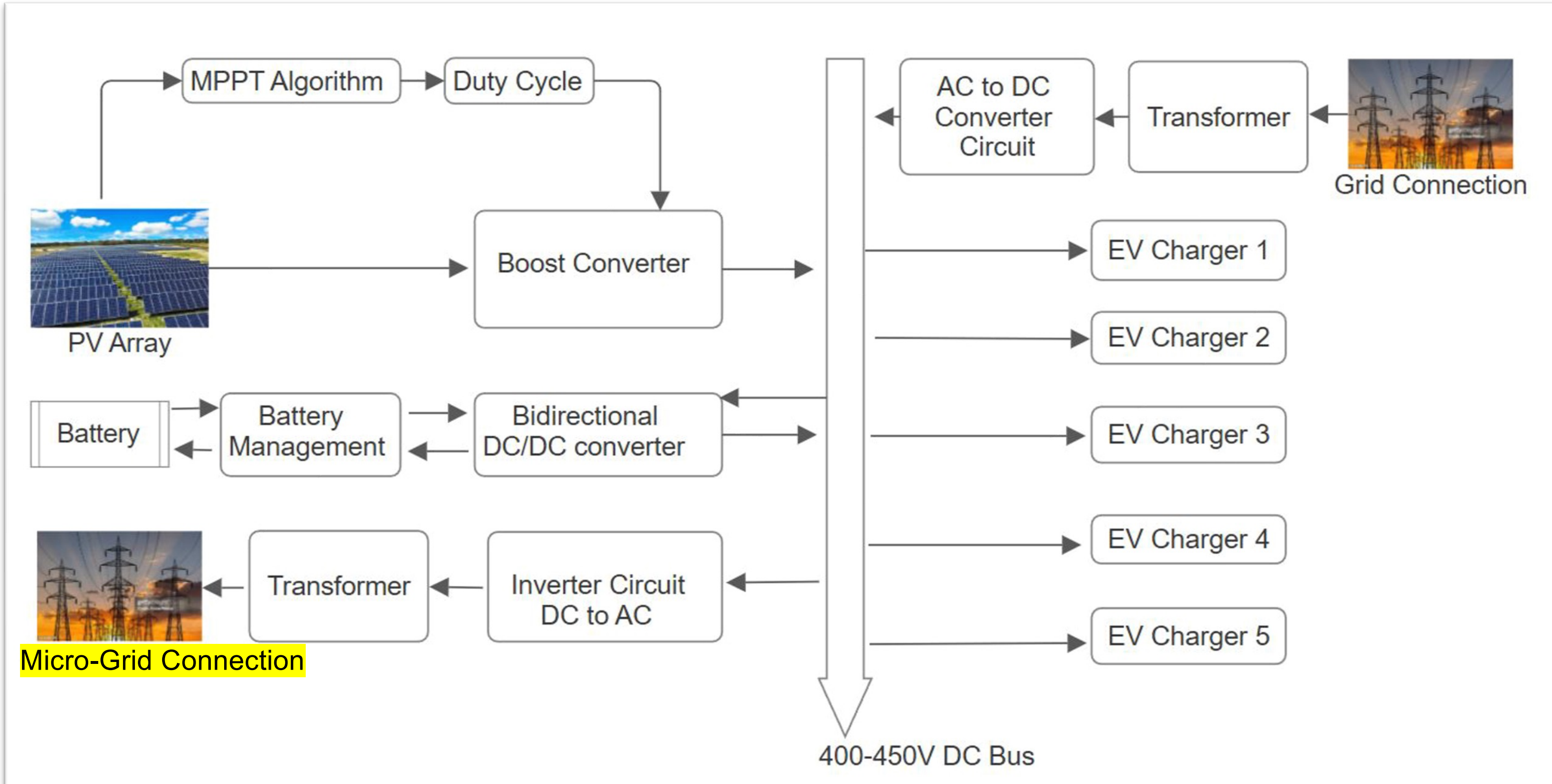
- Load Assessment
- System Specifications
- EV Charging System Design
- Energy Management System (EMS)
- Power Flow Simulation
- Microgrid Integration
- Battery Storage Optimization
- Solar System Optimization
- Material Selection
- Financial and Environmental Impact

Technical Specifications (Proposed) - |



Technical Specifications (Proposed) - ||

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- Load Analysis (DC – Fast Charger)
- Solar to DC Bus with MPPT Controller
- AC Grid to DC Bus through 3-Phase Rectifier
- DC Bus to Battery System Controlled by Battery Management System (BMS)
- DC Bus to DC Load (EV)
- DC Bus to Microgrid

Project Accomplishment - |

Our Team designed and simulated a **Smart EV Charging** station integrated with **Solar PV, Battery Storage (BESS)**, and the **Electrical Grid** to support sustainable transportation in remote areas like Jasper.

Key achievements include:

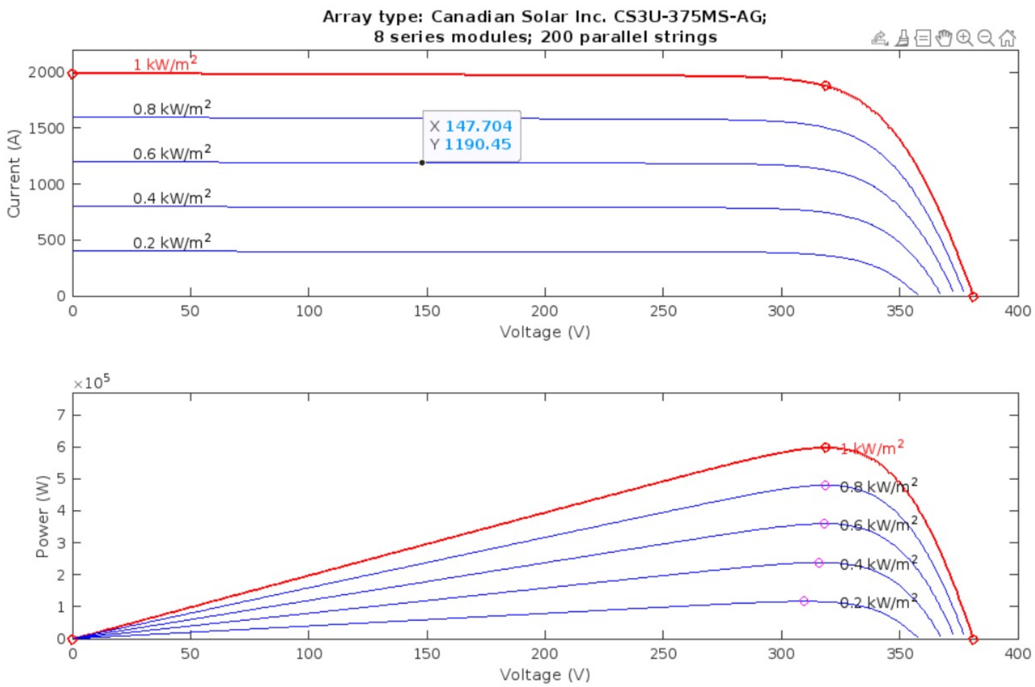
- Developed a **600kW solar PV system** with a **fuzzy logic-based MPPT controller** to maximize power extraction under varying conditions.
- Designed a **400V DC bus** for stable power distribution, fed by solar, grid, and a **550 kWh BESS**.
- Implemented a **Battery Management System (BMS)** with SOC-based control to optimize charging/discharging.
- Simulated **five 150 kW DC fast chargers** with **CCCV (Constant Current Constant Voltage) charging logic**.
- Created an **Energy Management System (EMS)** to prioritize solar → battery → grid power flow.
- Validated the system in **MATLAB/Simulink**, ensuring stability under different irradiance and load conditions.

Project Accomplishment - ||

Final Technical Specifications:

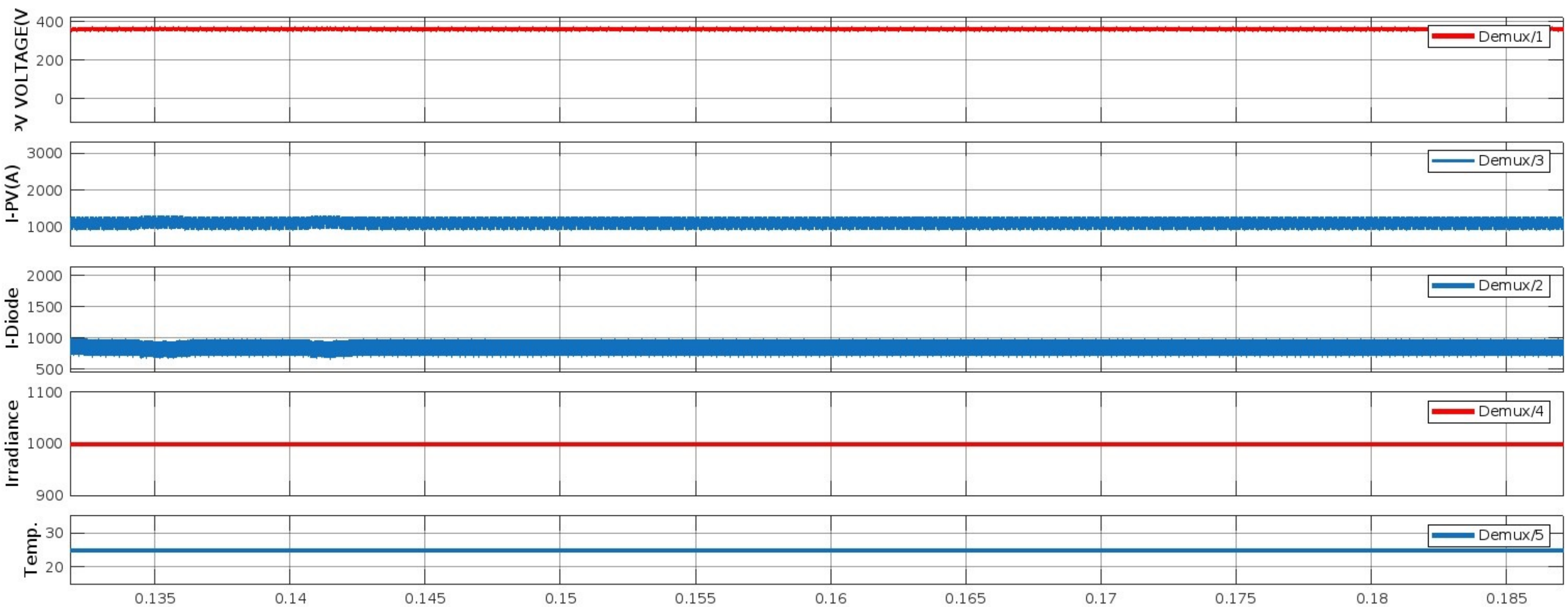
Component	Specification
Solar PV Array	600 kW (200 parallel strings × 8 series modules, Canadian Solar CS3U-375MS-AG).
MPPT Controller	Fuzzy logic-based, adaptive duty cycle scaling.
Battery (BESS)	550 kWh Li-ion, 400V nominal, 1400 Ah capacity, SOC-based control.
DC Bus Voltage	400V (±5%) stabilized by boost converter.
EV Chargers	5 × 150 kW, CCCV protocol (280A max current, 453V cutoff).
AC-DC Rectifier	3-phase PWM-controlled (208V AC → 400V DC).
DC-AC Inverter	3-phase, 400V DC → 208V AC for microgrid.
EMS Logic	Solar > Battery > Grid priority, load balancing.

Success and Validations - |



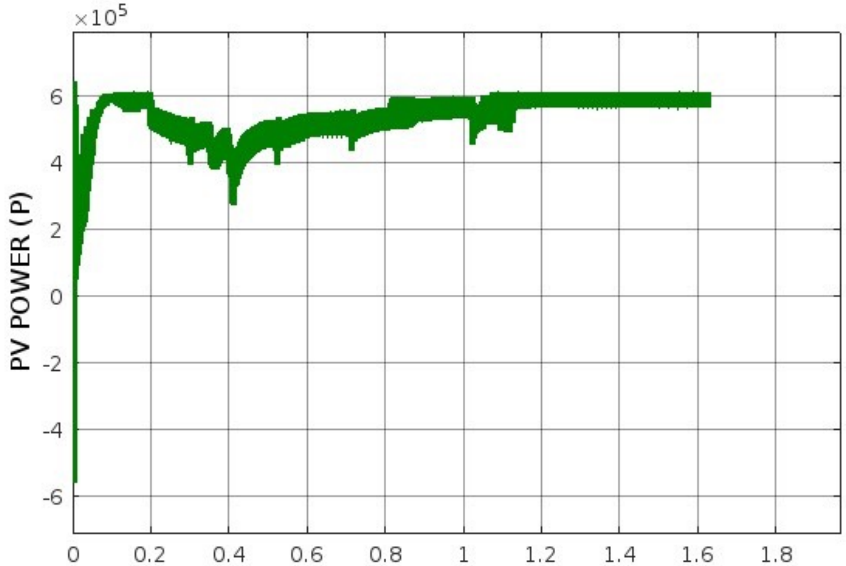
- In this figure, the power and current output at different irradiance have been shown at the maximum power point (MPP). We can observe that 1000 watts/m2, the power output is approximately 600 kW, and current is 2000A.

Success and Validations - ||

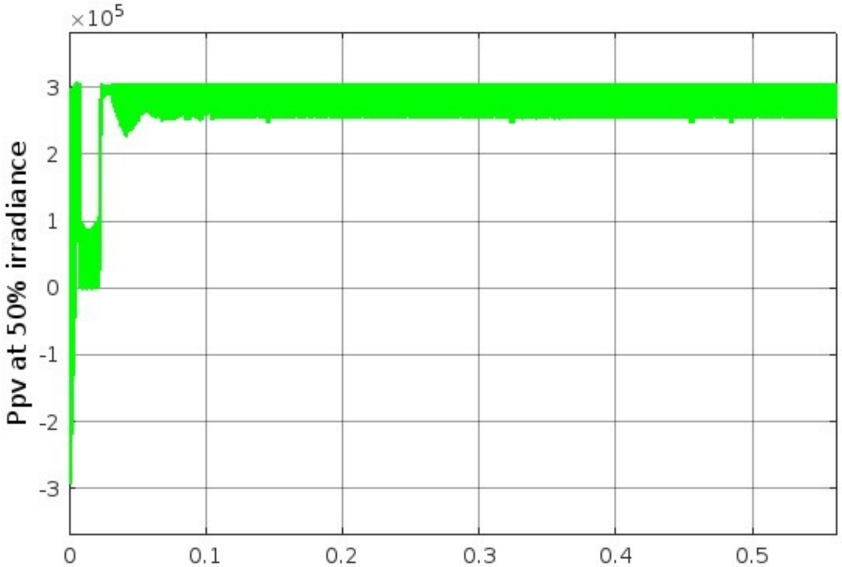


- The figure shows, the output voltage of PV, the output current of PV, the Current in diode, and a specific irradiance of 1000 watt/m2 and temperature of 25 deg-Celsius.

Success and Validations - | | |

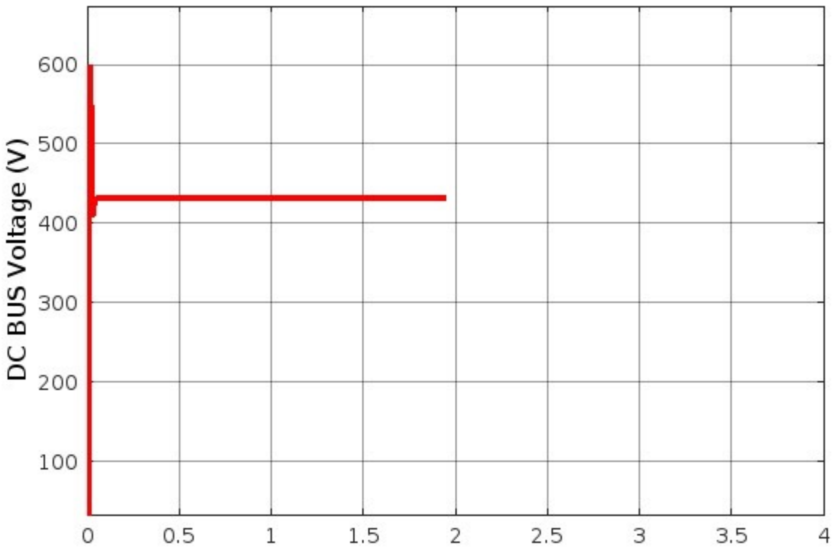


- This figure shows the PV power output at 1000 Watt/m2 irradiance.
- We observe the power output is around 580 kW, which was desired based on the proposal.



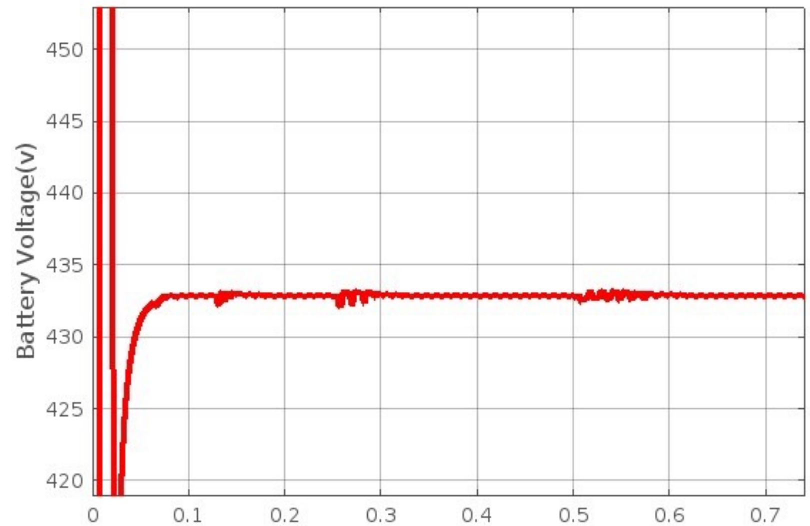
- This figure, demonstrates the power output reduction due to less irradiance (50%), The output consequently became around 300 kW, which indicates the PV output relation with irradiance.

Success and Validations - IV



- In this, we have achieved the DC BUS voltage of the system after the boost converter as 420 V, which is within the limit of the fixed DC bus voltage for charging the EV vehicles.

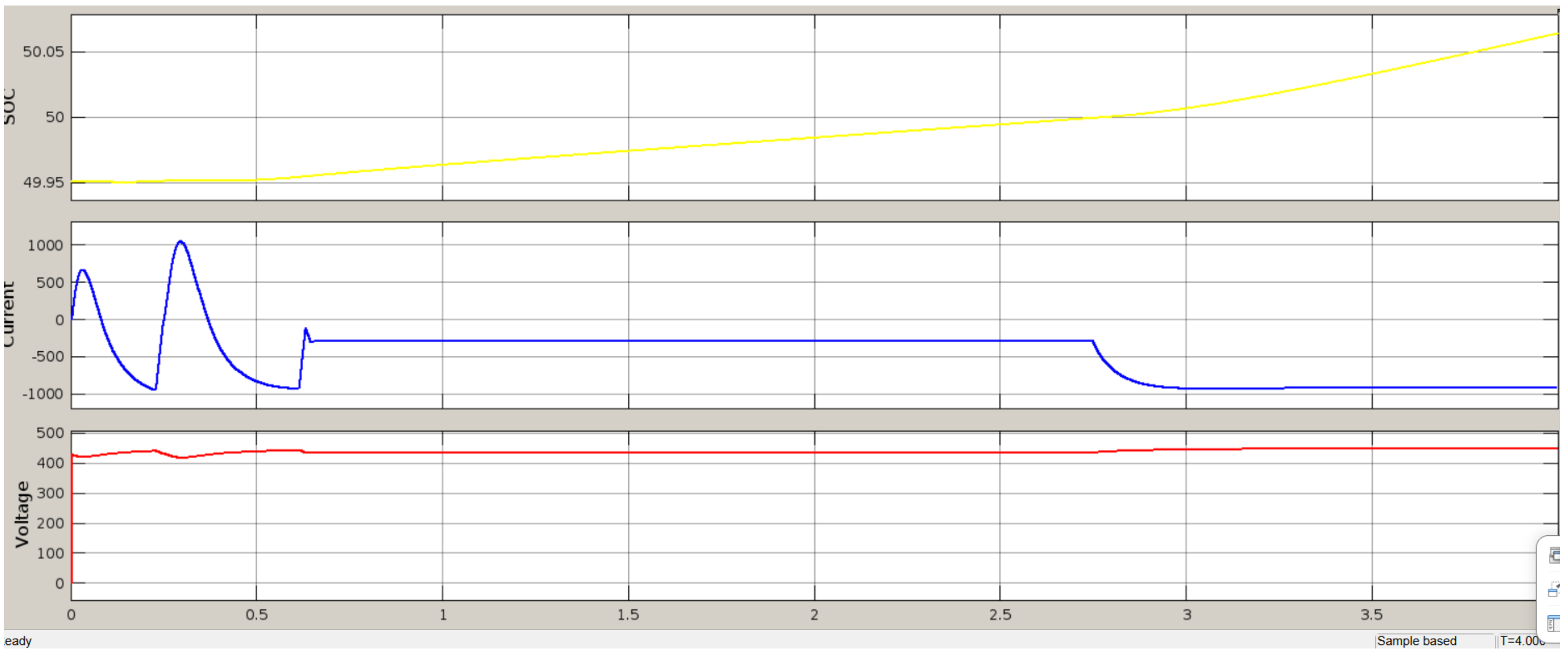
- The EV charging station is connected to this bus to keep the bus voltage constant to 400V +-5% is a validation of the system throughout the simulation.



- The Battery Output voltage is designed such that it maintains stability with DC bus voltage, as it will be synchronized with the DC bus and charge and discharge to that node.

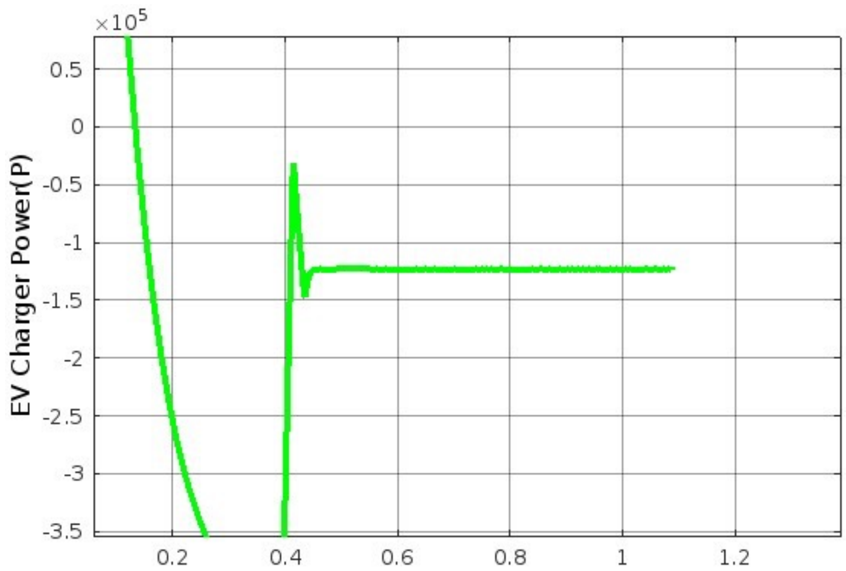
- We can see the DC battery voltage is approximately to a similar value of 420 V in the Dc bus.

Success and Validations - V

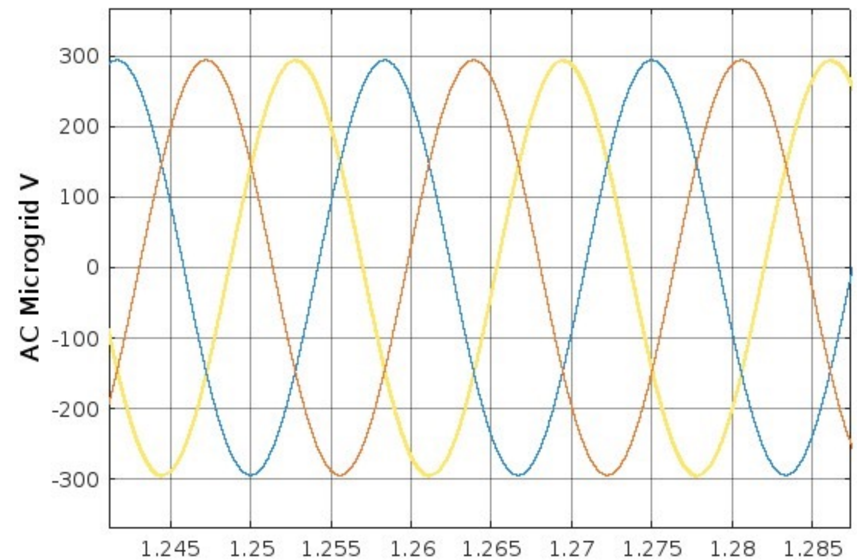


- The figure indicates the EV charger battery following the Constant current constant voltage method.
- The current has been set up to be charged at a maximum rate of 280 A (fixed based on the controller) to overcome over charging process. The voltage was kept constant at 400-450 volts to prolong battery life.

Success and Validations - VI

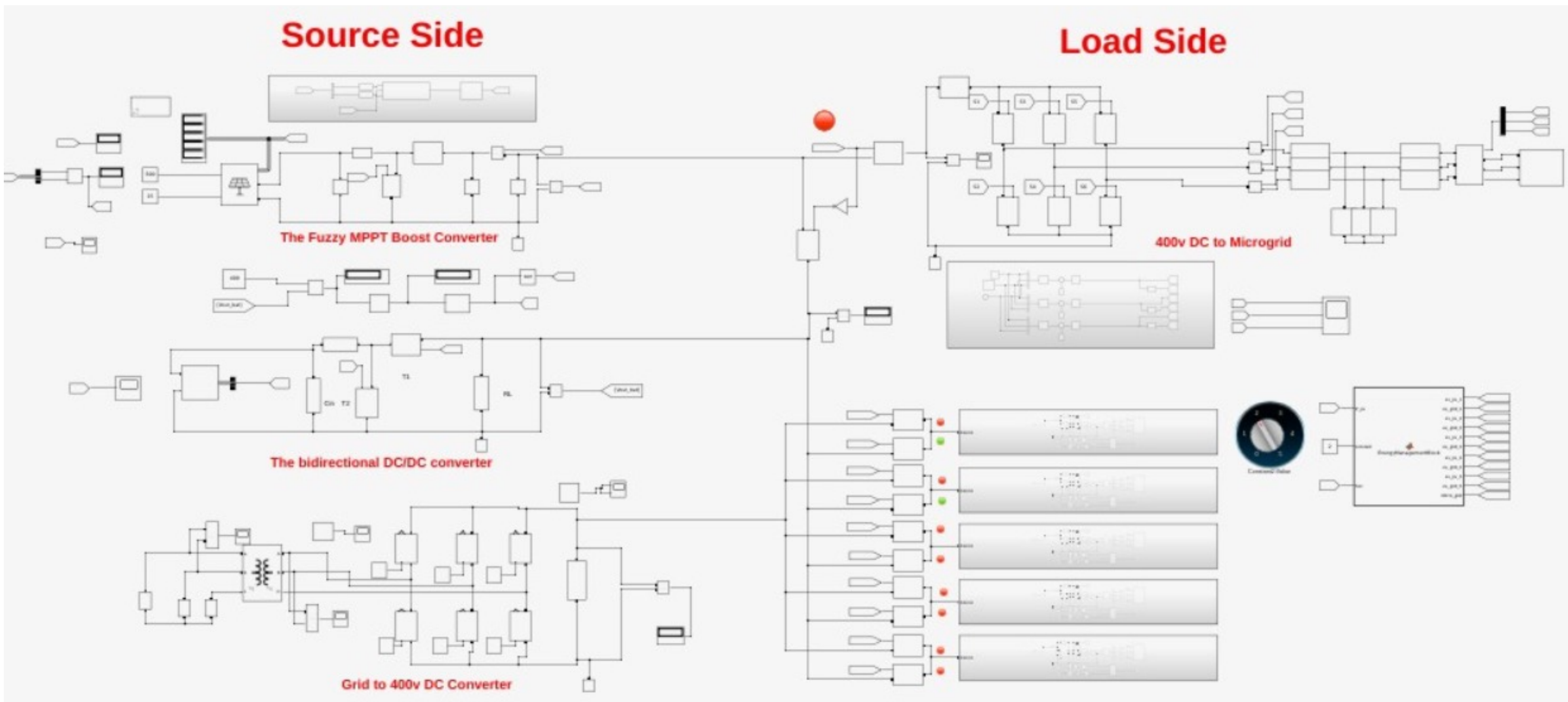


The figure shows the EV charger power for a single charger, which indicates charging at around 120-130kW.



- The figure indicates the AC Grid voltage in the microgrid side.
- The required Voltage is 3-phase 208 V is achieved after the 3-phase inverter circuit, that converts the DC voltage to AC voltage.

Success and Validations – VII



Deviations from proposal

1. The protection circuit design requires further elaboration and detailed implementation.
2. Short circuit analysis has been identified as a prospective area for future work.
3. The Battery Management System (BMS) can be improved by integrating additional parameters such as Battery Health and Depth of Discharge (DOD).
4. The current EV charger protocol is primarily based on logical design rather than practical application.

** AI Generated Image – Microsoft Designer